ES (E-mini S&P 500) — Engine Analysis Readout

Fund-manager assessment of the SS engines applied to ES. SEPARATE from the NQ work.

Points-based. ES = $50/point. v9-15m: 133 days (Jan–Jun 2026). v4-5m chop-filter: 46 days (May–Jun).

Bottom line up front

The engines port to ES with a positive but THIN edge on v9-15m, and a negative edge on v4-5m. ES is a fundamentally different instrument from NQ — lower price (~7,477 vs ~30,000), much smaller bar ranges (7pt vs 39pt on 15m), lower volatility. The engines' signal logic survives the transfer, but the per-trade edge shrinks to a level where costs become the deciding factor.

Headline numbers (exit-symbol, points)

	v9-15m	v4-5m (chop)
Days	133	46
Trades	354	1,833
Total points	**+388**	**−523**
Per-trade	+1.10	−0.29
Win rate	36%	34%
Win/loss size ratio	1.92	1.63

v9-15m is the one that works on ES — modestly positive (+388 pts, +1.10/trade), carried by a healthy 1.92 win/loss asymmetry (wins avg +29, losses avg −15) that overcomes the 36% hit rate. **v4-5m loses on ES** (−523, −0.29/trade) — the higher-frequency 5m engine generates too many small chop trades that ES's tight ranges can't pay for.

The cost problem (the fund manager's first concern)

v9's +1.10 pts/trade = **$55/trade gross** at $50/point. That sounds fine until you subtract:

- Commission: ~$4-5/round-turn on ES futures
- Slippage: ES is liquid, but a 5m/15m signal fill still slips ~0.25-0.5 pt = $12-25

Net, v9's $55/trade gross likely becomes ~$25-35/trade after costs — still positive, but the margin is thin and fragile. **v4-5m at −0.29/trade gross is dead on arrival after costs** — 1,833 trades each leaking commission is a guaranteed bleed.

The asymmetry matters: v9 takes 354 trades, v4 takes 1,833. On ES specifically, the low-frequency engine is far better positioned because each trade must clear a fixed cost, and ES's small point-moves give less room to do so.

Three-session breakdown (the standard model)

v9-15m:

Session	Trades	Points	Read
Evening (6–11:59PM)	193	**+826**	The profit engine — same as NQ
Overnight (12–9:30AM)	88	−54	~Flat, mild bleed
RTH (9:30–16:00)	68	**−280**	The weak spot — RTH loses

v4-5m (chop):

Session	Trades	Points
Evening	498	+158
Overnight	775	−195
RTH	515	−395

Same structural signature as NQ: evening is the profit center, RTH is the leak. On ES this is even more pronounced for v9 — its entire +388 comes from the evening (+826), while RTH actively loses (−280). The overnight is roughly flat. **An ES strategy that traded v9 only in the evening session and sat out RTH would likely outperform trading all day** — the same per-session insight from the NQ work applies here.

Consistency & risk

v9-15m monthly: Jan −61, Feb +188, Mar +212, Apr +106, May −102, Jun +44 → **4 of 6 months positive.** Reasonable consistency, but the two down months (−61, −102) show it's not all-weather. The edge is real but modest, and clusters in trending months.

Data quality note — material: 2 anomalous bars in the v9 feed (03-23 a 242-pt range, 04-07 a 120-pt range vs 7-pt median, same glitch type as the NQ feed). These are NOT negligible here: the 6 trades touching those dates contributed **+104 pts** — so v9's clean total is **+284, not +388**. The glitches happened to *help*. Scrubbed, v9 is +284 over 133 days (+0.80/trade), which makes the cost problem below even tighter. Always scrub these before any production read.

Fund-manager verdict

v9-15m on ES: a marginal-but-real edge, tradeable only with discipline. +284 pts/133 days (clean) is a positive expectancy, but at ~+0.80 pts/trade it lives or dies on execution costs. I would NOT trade it all-day. I WOULD consider it as an **evening-session-only** strategy (where the +826 evening profit lives), which concentrates the edge and cuts the RTH bleed. Even then it's a modest, cost-sensitive strategy, not a core alpha source.

v4-5m on ES: not viable. −523 pts and negative per-trade before costs. ES's tight ranges don't support the 5m engine's high trade frequency. The chop filter (carried over from NQ) isn't enough — ES needs either a much higher signal threshold or simply isn't suited to this engine at 5m.

The broader read: these engines were built and tuned on NQ. They **transfer** to ES (the signal logic produces a positive v9 evening edge) but they don't transfer *efficiently* — the edge compresses because ES is lower-vol. If ES is a target market, the engines would need re-tuning to ES's volatility (wider relative thresholds, evening-session focus, and for v4 a much heavier frequency filter or a move to a higher timeframe). As-is, only v9's evening session clears the bar, and only marginally.

Caveats

- v9 has 133 days (decent); v4 only 46 days (thin — treat its −523 as indicative, not conclusive).
- These are gross-of-cost point sims with next-bar-open fills. Real ES costs would meaningfully erode v9 and bury v4.
- ES analysis is fully separate from the NQ research — no thresholds or findings carried over.